Cobweb Forests:
A Kolmogorov–Arnold Theory of Networks of Concept Hierarchies

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Abstract

The Kolmogorov–Arnold (KA) representation theorem states that every continuous function of n variables is a finite superposition of continuous functions of one variable and addition, and Kolmogorov–Arnold Networks (KANs) turn that statement into an architecture with learnable univariate functions on its edges. We ask the two analogous questions for Cobweb, the incremental concept-formation system whose category utility is the mutual information between features and concepts. *Representation:* does there exist a set of Cobweb trees that represents an arbitrary continuous function? *Learnability:* does Cobweb's own learning procedure—category utility and its four operators, with no gradients—learn it? We formalise a *Cobweb forest*: a directed acyclic graph of hybrid Cobweb trees in which each tree reads raw features and the *messages* of other trees—their *activations* (multi-node predictions of a continuous attribute) and their *labels* (the concepts an input sorts to)—and one designated tree produces the output. We prove (i) that a single tree is a universal approximator under the prediction rules of the hybrid implementation, but that the prior-weighted mixture over all concepts is not consistent; (ii) a Cobweb–Kolmogorov–Arnold theorem: forests whose trees have at most two inputs and pass activations are universal, with $4n^2 + 4n$ trees, and two inputs per tree are necessary; (iii) that label forests are universal as well; (iv) a structural invariant of Cobweb's operators (no leaf is ever destroyed), which makes a single learned tree a universally consistent regressor independently of the structure category utility builds; and (v) that a forest learned by Cobweb alone is consistent with a rate set by the trees' fan-in rather than by n whenever its messages are *sufficient*, while activation messages provably fail on functions such as xy and label messages repair them. Each statement is paired with an experiment in the style of the KAN paper.

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1 Introduction

Kolmogorov–Arnold. For every $n \geq 2$ there are continuous functions $\varphi_{q,p} : [0,1] \to \mathbb{R}$ ($q = 0, \ldots, 2n$, $p = 1, \ldots, n$) such that every $f \in C([0,1]^n)$ can be written as

$$f(x_1, \ldots, x_n) = \sum_{q=0}^{2n} \Phi_q\Big(\sum_{p=1}^{n} \varphi_{q,p}(x_p)\Big) \tag{1}$$

with continuous outer functions $\Phi_q : \mathbb{R} \to \mathbb{R}$ that depend on f [1, 8, 10, 14]. The only genuinely multivariate operation is addition. The inner functions may be very rough (nowhere differentiable), which long made (1) look irrelevant to learning [6]. Kolmogorov–Arnold Networks [9] stack layers of learnable univariate functions (B-splines on edges, sums at nodes) and prove an approximation theorem whose rate does not depend on n when f has a *smooth* KA-type representation [9, Thm. 2.1].

Cobweb. Cobweb [5] sorts each instance down a hierarchy of probabilistic concepts and, at every node, applies whichever of four operators—add to the best child, create a new child, merge the two best children, split the best child—most increases category utility, which in its information-theoretic form is the mutual information between features and concepts [2, 12]. Concepts store sufficient statistics (attribute–value counts for nominal attributes, a mean and variance for each continuous one), so learning is closed-form, incremental and does not forget. Prediction expands concepts best-first and mixes their predictions [12].

Several recent systems use Cobweb trees as *modules* whose representations become the inputs of other trees: Convolutional Cobweb [11] sorts image patches into a filter hierarchy and describes each image by the patches' filter labels for a second, classification hierarchy; TRELLIS [13] describes composites by bags of concept identifiers drawn from another hierarchy; and a Cobweb language model passes each tree's prediction for a position to the trees above it. We call such a network a *Cobweb forest* and ask what it can represent and learn.

Questions.

Q1. *Representation.* For every continuous f and $\varepsilon > 0$, is there a Cobweb forest whose output is uniformly within ε of f? How few trees, and how few inputs per tree, suffice?

Q2. *Learnability.* If every tree learns only with Cobweb's own procedure, does the forest's output converge to f? At what rate, and what decides it?

Contributions.

- A formal model of the hybrid Cobweb tree as implemented in `cobweb_hybrid`, including its four prediction weightings, and of Cobweb forests with activation and label messages (Section 2).

- Representation (Section 3): single-tree universality with an explicit rate (Theorem 9); inconsistency of the prior-weighted all-concept mixture (Proposition 10); the Cobweb–Kolmogorov–Arnold theorem for fan-in-two activation forests (Theorem 12) and the necessity of fan-in two (Proposition 13); universality of label forests (Theorem 15); and the observation that merging ("interweaving") trees under a common root yields mixtures, not compositions (Proposition 16).
- Learnability (Section 4): the leaf-preservation invariant of Cobweb's operators (Lemma 17); universal consistency of a learned tree, independent of the structure category utility chooses (Theorem 18); consistency of Cobweb-learned forests under a sufficiency condition, at a rate governed by fan-in (Theorem 21); the failure of activation messages without sufficiency (Proposition 23); and consistency of unsupervised label forests as data-dependent partitioning estimates (Theorem 24).
- A test battery that mirrors the KAN paper (Section 5).

2 The model

2.1 Concepts and trees

An *instance* is a pair $z = (a, u)$ of a nominal part a (a table that assigns to each attribute a bag of values with weights) and a continuous part $u \in \mathbb{R}^m$.

Definition 1 (Concept)**.** A concept c stores its count n_c, the attribute–value counts $n_c(A{=}v)$ of the instances it covers, and, for the continuous block, their mean $\mu_c \in \mathbb{R}^m$ and empirical (per-dimension) variance $s_c^2 \in \mathbb{R}^m_{\geq 0}$. With smoothing parameters $\alpha > 0$ and $h^2 > 0$ (`alpha`, `prior_var`) its model is

$$P_c(A{=}v) = \frac{n_c(A{=}v) + \alpha}{n_c(A) + (K_A + 1)\alpha}, \qquad p_c(u_j) = \mathcal{N}\big(u_j;\, \mu_{c,j},\, v_{c,j}\big), \quad v_{c,j} = s_{c,j}^2 + h^2,$$

with attributes conditionally independent given c, K_A the number of values of A seen so far.

Definition 2 (Cobweb tree)**.** A Cobweb tree $\mathcal{T}$ is a rooted tree of concepts whose statistics are *additive*: each internal concept's statistics are those of the pooled instances of its children. Its *category utility* at a node with children C is the normalised information gain $\mathrm{PU} = |C|^{-1}\big[H(\text{parent}) - \sum_{c \in C} P(c) H(c)\big]$, where H is the sum of the Dirichlet-smoothed categorical entropies and of the Gaussian differential entropies $\frac{1}{2}\log(2\pi e\, v_{c,j})$; without the $|C|^{-1}$ factor this is the mutual information $I(Z; C)$ [2, 12].

Learning (`ifit`) sorts an instance from the root. At a leaf it either increments the leaf (exact match) or performs a *fringe split*: the leaf becomes an internal node with the old leaf and a new leaf for the instance as children. At an internal node it adds the instance and chooses, by category utility, between BEST (descend into the best child), NEW (create a leaf), MERGE (join the two best children under a new node) and SPLIT (replace the best child by its children).

2.2 Prediction

A *query* is an instance in which some attributes and some continuous coordinates are missing; we write O for the observed coordinates (missing continuous coordinates are encoded as `NaN`, missing attributes are absent). The log-likelihood of a query under a concept is $\ell_c(q) = \sum_{A\ \text{obs.}} \log P_c(A{=}q_A) + \sum_{j \in O} \log p_c(q_j)$.

Definition 3 (Multi-node prediction)**.** Prediction expands concepts best-first from the root (at most M of them), scoring each concept c with one of

$$\begin{aligned}
&\text{LIK}: && s_c = \ell_c(q),\\
&\text{PRIOR}: && s_c = \ell_c(q) + \log(n_c/n_{\text{root}}),\\
&\text{PATH}: && s_c = \pi_c(q) + \ell_c(q), \qquad \pi_c(q) = \textstyle\sum_{c' \in \text{path}(c)\setminus\{\text{root}\}} \log \frac{n_{c'} e^{\ell_{c'}(q)}}{\sum_{b \in \text{sib}(c')} n_b e^{\ell_b(q)}},\\
&\text{POSTERIOR}: && s_c = \pi_c(q),
\end{aligned}$$

where $\text{sib}(c')$ are the children of c''s parent (including c'). With $E_M(q)$ the expanded set and $w_c(q) \propto e^{s_c}$ normalised over $E_M(q)$, the prediction of a missing continuous coordinate j and of a missing attribute A is

$$\hat{y}_j(q) = \sum_{c \in E_M(q)} w_c(q)\, \mu_{c,j}, \qquad \hat{P}(A{=}v \mid q) = \sum_{c \in E_M(q)} w_c(q)\, P_c(A{=}v).$$

The *leaves-only* variant restricts the mixture to the expanded leaves.

PATH is the default of the implementation and of `cobweb_torch`; $\pi_c(q)$ is the log posterior of c obtained by multiplying the sibling-normalised posteriors down its path, so $\sum_{c \text{ at depth } k} e^{\pi_c} \leq 1$ for every depth k. The rational account [12] ranks concepts by pointwise mutual information $\log p(x \mid c) - \log p(x)$, i.e. by ℓ_c.

2.3 Cobweb forests

Definition 4 (Messages)**.** Let $\mathcal{T}$ be a tree whose continuous block contains *task* coordinates $Y_{\mathcal{T}}$ (possibly none). For a query q in which $Y_{\mathcal{T}}$ is missing:

- the *activation* of $\mathcal{T}$ is its prediction $\hat{y}_{Y_{\mathcal{T}}}(q) \in \mathbb{R}^{|Y_{\mathcal{T}}|}$;
- the *label* of $\mathcal{T}$ at depth k is the identifier of the depth-k concept on the root-to-leaf descent that chooses, at each node, the child maximising $\log n_c + \ell_c(q)$ (the depth of a leaf if the path is shorter; $k = \infty$ denotes the leaf).

Definition 5 (Cobweb forest)**.** A Cobweb forest $\mathfrak{F} = (V, E, o)$ is a finite directed acyclic graph whose vertices are Cobweb trees and whose edges $\sigma \to \tau$ carry the activation and/or the label of σ into τ, together with an output tree $o \in V$. A tree τ reads an instance made of (a) the raw coordinates x_{R_τ} it is wired to, (b) the activations of its parents as continuous coordinates, (c) the labels of its parents as nominal attributes (one attribute per parent), and (d) its task attributes, if any. For an input x the forest computes the messages of every tree in topological order and outputs the activation (regression) or the predicted value distribution (classification) of o. The *fan-in* of τ is the number of its continuous non-task inputs plus the number of its label inputs.

Definition 6 (Learning a forest with Cobweb)**.** Given a stream $(x_i, y_i)_{i \leq N}$, every tree learns only with `ifit` on the instances defined in Definition 5; a tree's task attributes, if any, are the target y (the target "funnels" through that tree), and a tree without task attributes is unsupervised. Two protocols define which messages a tree is trained on:

- *online*: for each i, every tree first emits its message for x_i from what it has learned from $(x_{i'})_{i'<i}$, then learns instance i (messages are out-of-sample, as at test time);
- *split*: the stream is cut into consecutive chunks, one per level of the DAG; a tree at level k learns chunk k with the messages of its already-trained (frozen) parents.

Remark 7 (Why not train stage-wise on the same data). A tree that has learned (x_i, y_i) keeps a leaf for it (Lemma 17), and its activation at x_i is close to y_i itself. A parent trained on such in-sample messages learns the relation between y and a leaked copy of y, which it never sees at test time. The online protocol (predict, then learn), already used by the Cobweb language model, and the split protocol avoid this; the experiments report the leaking protocol as a negative control.

3 Representation

Throughout, a tree that *represents* $g : K \to \mathbb{R}$ ($K \subset \mathbb{R}^d$ compact) is a tree over instances $(x, g(x))$ queried with the last coordinate missing, and $\omega_g(r) = \sup\{|g(x) - g(x')| : \|x - x'\| \le r\}$ is the modulus of continuity.

3.1 A local error certificate

Lemma 8 (Concept bias). *Let c be a concept of a tree over instances $(x_i, g(x_i))$ with g L-Lipschitz. Then $|\mu_{c,Y} - g(\mu_{c,X})| \le L\sqrt{\operatorname{tr} S_c}$, where S_c is the empirical covariance of the x-part of c's instances, and consequently for every query x*

$$|\hat{y}(x) - g(x)| \;\le\; L \sum_{c \in E_M(x)} w_c(x)\Big(\|x - \mu_{c,X}\| + \sqrt{\operatorname{tr} S_c}\Big). \tag{2}$$

Proof. $|\mu_{c,Y} - g(\mu_{c,X})| = |\frac{1}{n_c}\sum_i (g(x_i) - g(\mu_{c,X}))| \le \frac{L}{n_c}\sum_i \|x_i - \mu_{c,X}\| \le L(\frac{1}{n_c}\sum_i \|x_i - \mu_{c,X}\|^2)^{1/2}$ by Jensen. For (2), write $\hat{y} - g(x) = \sum_c w_c(\mu_{c,Y} - g(\mu_{c,X})) + \sum_c w_c(g(\mu_{c,X}) - g(x))$. □

The right side of (2) is computable from the tree; it says that a mixture is accurate exactly when it puts its weight on concepts that are near the query *and* tight. Every result below is a way of guaranteeing that.

3.2 One tree

Theorem 9 (A single tree is a universal approximator). *Let $g \in C(K)$, $K \subset \mathbb{R}^d$ compact, and $\varepsilon > 0$. There is a Cobweb tree $\mathcal{T}$ and a variance floor $h^2 > 0$ such that $\sup_{x \in K} |\hat{y}_{\mathcal{T}}(x) - g(x)| < \varepsilon$ under full expansion ($M = \infty$) with each of the weightings* LIK *and* PATH*, and with the leaves-only variant of any weighting. If g is L-Lipschitz, a tree with G leaves achieves*

$$\sup_{x \in K} |\hat{y}_{\mathcal{T}}(x) - g(x)| \;\le\; C_d\, L\, G^{-1/d}\sqrt{\log G} + C_d \|g\|_\infty G^{-1}. \tag{3}$$

Proof. Extend g continuously to a cube $Q \supseteq K$ (Tietze) and rescale $Q = [0,1]^d$. Let $\delta = \beta h$ with a fixed $\beta > 0$ and let Γ be the grid of cell centres of spacing δ, $G = |\Gamma| = \delta^{-d}$. Take the depth-one tree whose leaves are the instances $(\gamma, g(\gamma))$, $\gamma \in \Gamma$, and whose root pools them. Each leaf has $s^2 = 0$, hence variance h^2 in every coordinate; the root has variance at least $v_0 > 0$ (the variance of the grid) in every coordinate. Write $K_h(r) = (2\pi h^2)^{-d/2} e^{-r^2/2h^2}$ and $k_\gamma = K_h(\|x - \gamma\|)$.

Leaves dominate. Some γ lies within $\sqrt{d}\,\delta/2$ of x, so $\max_\gamma k_\gamma \ge (2\pi h^2)^{-d/2} e^{-d\beta^2/8}$, while $\sum_\gamma k_\gamma \le (2\pi h^2)^{-d/2} C(\beta, d)$ by comparison with a Gaussian sum over $\delta\mathbb{Z}^d$. Under LIK the leaves carry unnormalised mass $\sum_\gamma k_\gamma \ge c\, h^{-d}$ and the root at most $(2\pi v_0)^{-d/2}$. Under PATH the leaf γ has $e^{\pi_\gamma} = k_\gamma / \sum_{\gamma'} k_{\gamma'}$ (equal counts), so its mass is $k_\gamma^2 / \sum k$, whose sum is at least $(\max k)^2 / \sum k \ge c' h^{-d}$, while the root again has mass $O(1)$. In both cases the root's weight is $O(h^d) = O(G^{-1})$.

Leaves form a kernel smoother. Among the leaves, $w_\gamma \propto k_\gamma^r$ with $r = 1$ (LIK) or $r = 2$ (PATH), i.e. a normalised Gaussian kernel of bandwidth $h/\sqrt{r}$ on the grid. For any $\rho > 0$,

$$\Big|\sum_\gamma w_\gamma g(\gamma) - g(x)\Big| \le \omega_g(\rho) + 2\|g\|_\infty \sum_{\|\gamma - x\| > \rho} w_\gamma \le \omega_g(\rho) + 2\|g\|_\infty C'' e^{-\rho^2/2h^2},$$

using the same lower bound on the normaliser. Taking $\rho = \sqrt{h}$ makes the bound vanish as $h \to 0$, uniformly in x; adding the root's $O(h^d)$ weight (times $2\|g\|_\infty$) gives the first claim. For Lipschitz g take $\rho = h\sqrt{2\log(1/h)}$, so $\omega_g(\rho) \le L\rho$ and the tail is $O(h)$; with $h \asymp G^{-1/d}$ this is (3). The leaves-only variant drops the root term. □

The PRIOR weighting behaves differently: it multiplies each concept's likelihood by its *relative count*, and the internal concepts of a tree then keep a share of the mixture that does not vanish as the variance floor shrinks.

Proposition 10 (The prior-weighted all-concept mixture is inconsistent)**.** *In the setting of the proof of Theorem 9, with PRIOR weighting and full expansion, the root's weight $\lambda(x)$ satisfies $\lambda(x) \to \mathcal{N}(x; \mu_{root}, v_{root})/(1 + \mathcal{N}(x; \mu_{root}, v_{root})) \in (0, 1)$ as $h \to 0$, so $\hat{y}(x) \to (1 - \lambda(x))g(x) + \lambda(x)\bar{g}$ with $\bar{g}$ the mean of g over the grid: the bias does not vanish unless g is constant. The same holds in any tree in which some internal concept $c \ni x$ keeps n_c/n_{root} and v_c bounded away from zero.*

Proof. Leaf γ has prior $1/G$, so the leaves' mass is $G^{-1}\sum_\gamma k_\gamma$, a Riemann sum of the Gaussian kernel density estimate of the uniform density on $[0,1]^d$, which tends to 1 for x in the interior. The root has prior 1 and mass $\mathcal{N}(x; \mu_{\text{root}}, v_{\text{root}})$, which does not depend on h in the limit. □

Proposition 10 explains the flat error of the prior weighting in the sine experiment of Section 5 (RMSE $0.20 \to 0.14 \to 0.12$ while PATH falls $0.034 \to 0.015 \to 0.005$ and leaves-only LIK $0.013 \to 0.006 \to 0.0015$, for $N = 100, 400, 1600$).

3.3 Composition: the Cobweb–Kolmogorov–Arnold theorem

Lemma 11 (Approximating a DAG of continuous maps)**.** *Let D be a finite DAG of continuous maps $g_v : \mathbb{R}^{k_v} \to \mathbb{R}$ whose inputs are raw coordinates of $x \in [0,1]^n$ or outputs of other vertices, and let F be the map it computes; let K_v be the (compact) set of input vectors of v as x ranges over $[0,1]^n$. For every $\varepsilon > 0$ there are $\rho_v, \eta_v > 0$ such that any maps $\hat{g}_v$ with $\sup_{K_v + B_{\rho_v}} |\hat{g}_v - g_v| \le \eta_v$ define a network $\hat{F}$ with $\sup_x |\hat{F}(x) - F(x)| \le \varepsilon$.*

Proof. Process the vertices in reverse topological order. At the output vertex put $\eta_o = \varepsilon/2$ and choose $\rho_o \le 1$ with $\omega_{g_o}(\rho_o) \le \varepsilon/2$ on the compact $K_o + \bar{B}_1$. Given the tolerance ρ_v that the inputs of v may deviate by, split it among the incoming vertices u (each may deviate by $\rho_v/\sqrt{k_v}$) and set, for each such u, $\eta_u = \rho_v/(2\sqrt{k_v})$ and ρ_u with $\omega_{g_u}(\rho_u) \le \rho_v/(2\sqrt{k_v})$ on $K_u + \bar{B}_1$ (taking minima over the several children of a vertex). Induction in topological order shows that every vertex's input deviates from its exact value by less than ρ_v, so $|\hat{g}_v - g_v| \le \eta_v + \omega_{g_v}(\rho_v)$ at the computed inputs, and the output error is at most ε. □

Theorem 12 (Cobweb–Kolmogorov–Arnold)**.** *Let $n \ge 2$. For every $f \in C([0,1]^n)$ and $\varepsilon > 0$ there is a Cobweb forest with activation messages, in which every tree has at most two inputs, consisting of*

- *$n(2n+1)$ univariate trees for the inner functions $\varphi_{q,p}$ (independent of f),*

- $(2n+1)(n-1)$ *two-input trees forming the inner sums,*
- $2n+1$ *univariate trees for the outer functions* Φ_q*, and*
- $2n$ *two-input trees forming the outer sum,*

$4n^2+4n$ *trees in all, whose output is within* ε *of* f *on* $[0,1]^n$*.*

Proof. Write f in the form (1) and read it as a DAG of continuous maps of one or two variables: $\varphi_{q,p}$, the binary additions $(a,b)\mapsto a+b$ chained to form the $2n+1$ inner sums of n terms and the outer sum of $2n+1$ terms, and Φ_q. Apply Lemma 11, and realise each $\hat{g}_v$ on the compact $K_v+\bar{B}_{\rho_v}$ by Theorem 9 as a tree over instances $(\text{inputs}, g_v(\text{inputs}))$ whose task coordinate is the value of g_v; its activation is the required $\hat{g}_v$. The count follows from the structure of (1). □

Proposition 13 (Two inputs are necessary)**.** *If every tree of a forest has fan-in one, its output depends on a single raw coordinate. Hence fan-in two is the least fan-in at which Cobweb forests are universal.*

Proof. By induction in topological order, each tree's messages are functions of the one raw coordinate at the source of its unique input chain. □

Remark 14 (Rates and smooth KA representations)*.* Theorem 12 uses a fixed number of trees, but for an arbitrary f the inner functions of (1) are rough, and the number of leaves each tree needs is not controlled. As for KANs [9, Thm. 2.1], the useful statement assumes structure. If f is computed by a DAG of s maps of fan-in $\le k$, each L-Lipschitz on a neighbourhood of its input set, then combining (3) with the stability argument of Lemma 11 gives a forest of s trees with G leaves each and error $C(s,L)\,G^{-1/k}\sqrt{\log G}$—a rate set by the fan-in k, not by n. A single tree over the n inputs has rate $G^{-1/n}$ (Theorem 9). Note that Cobweb trees realise sums as learned two-input maps; KANs sum at nodes for free, which is why their rate is governed by one-dimensional splines.

3.4 Label forests

Theorem 15 (Label forests are universal)**.** *For every* $f\in C([0,1]^n)$ *and* $\varepsilon>0$ *there is a forest of* n *unsupervised univariate trees that pass leaf labels to an output tree whose instances consist of the* n *labels (nominal attributes) and the target, such that the output tree's prediction (*LIK*, leaves only, full expansion) is uniformly within* ε *of* f*.*

Proof. Let tree p be the depth-one tree over the K grid points of spacing $\delta=1/K$ in $[0,1]$, all with count one and variance h^2. Its label for x_p is $\arg\max_\gamma\{\log n_\gamma+\ell_\gamma(x_p)\}$, the nearest grid point, so the labels partition $[0,1]^n$ into boxes B_t of side δ, indexed by label tuples t. Let the output tree be the depth-one tree with one leaf per tuple t, holding the instance $(t, f(\text{centre}(B_t)))$. A query with tuple t gives leaf t' that disagrees with t in m attributes the likelihood ratio $\big(\alpha/(1+\alpha)\big)^m$ relative to leaf t (Definition 1, counts one). There are at most $\binom{n}{m}K^m$ such leaves, so their total weight is at most $(1+K\alpha)^n-1$. Hence $|\hat{y}(x)-f(x)|\le\omega_f(\sqrt{n}\,\delta/2)+2\|f\|_\infty\big((1+K\alpha)^n-1\big)$, which is below ε for K large and then α small. □

The label forest of Theorem 15 has an output tree with K^n leaves: labels carry no metric, so this construction pays for every cell of the product partition. Activations carry the metric and allow the fan-in-two construction of Theorem 12; a hybrid message (label and activation) carries both.

3.5 Interweaving trees

Proposition 16 (Merging trees under a common root gives mixtures)**.** *Let $\mathcal{T}_1, \ldots, \mathcal{T}_m$ be trees over the same instance space, and let $\mathcal{T}$ be the tree whose root pools their roots' statistics and has $\mathcal{T}_1, \ldots, \mathcal{T}_m$ as its subtrees. Under full expansion with* LIK *or* PATH *weighting,*

$$\hat{y}_{\mathcal{T}}(q) = \lambda_0(q)\,\mu_{root} + \sum_{j=1}^{m} \lambda_j(q)\,\hat{y}_{\mathcal{T}_j}(q), \qquad \lambda_j \geq 0,\ \sum_{j=0}^{m} \lambda_j = 1.$$

Proof. Under LIK a concept's score does not depend on the tree it sits in, so the mixture over $\mathcal{T}$ splits into the root term and the m subtree mixtures, with $\lambda_j \propto \sum_{c \in \mathcal{T}_j} e^{\ell_c}$. Under PATH every concept of $\mathcal{T}_j$ gains the same term $\log P(\mathcal{T}_j \mid q)$ in π_c, which rescales the subtree's weights uniformly and leaves its internal mixture unchanged. □

Interweaving therefore closes the class of tree predictors under input-dependent convex combinations; it never composes them. Composition—one tree reading another's representation—is what forests add.

4 Learnability

4.1 An invariant of Cobweb's operators

Lemma 17 (Leaf preservation)**.** *Run* `ifit` *on any sequence $z_1, \ldots, z_N$ with any choices of operators (including add-only training), and suppose the depth limit is never reached. Then (i) every concept's statistics are the pooled statistics of the instances stored in the leaves below it; (ii) no leaf is ever deleted; and (iii) the leaves are in bijection with the distinct instances, where two instances are the same if they agree up to the exact-match tolerance.*

Proof. By induction over the branches of the sorting loop. BEST increments the current node and descends; NEW increments the current node and creates a leaf holding z; MERGE inserts a new internal node whose statistics are the pooled statistics of the two children it adopts; SPLIT deletes the best child, which is internal (the operator requires it to have at least two children), and promotes its children; at a leaf, an exact match increments it and any other instance triggers a fringe split, which copies the leaf's statistics into a new internal parent, keeps the old leaf, and adds a leaf for z. No branch deletes a leaf, every instance ends in exactly one leaf, and a leaf only absorbs exact matches. □

When the depth limit is reached the implementation inserts the instance into the current node, which may be a leaf; (iii) then fails, which is why the lemma assumes the default limit (1000) is not reached.

4.2 One learned tree

Theorem 18 (A learned tree is universally consistent)**.** *Let (X_i, Y_i) be i.i.d. with $X \in \mathbb{R}^d$, $|Y| \leq B$, $m(x) = \mathbb{E}[Y \mid X = x]$. Let $\mathcal{T}_N$ be the tree that Cobweb learns from $(X_i, Y_i)_{i \leq N}$ in any order and with any operator choices, and let it predict Y from X with the leaves-only mixture,* LIK *weighting and full expansion, with variance floor h_N^2. If X has a density, $h_N \to 0$ and $N h_N^d \to \infty$, then $\int |\hat{y}_{\mathcal{T}_N}(x) - m(x)|\,\mu(dx) \to 0$ almost surely, whatever structure category utility builds (μ the distribution of X).*

Proof. With a density the X_i are distinct almost surely and so are the instances, so by Lemma 17 the tree has exactly N leaves, leaf i holding (X_i, Y_i) with zero empirical variance. The LIK score of leaf i for a query x with Y missing is $\log \mathcal{N}(x; X_i, h_N^2 I)$, so the leaves-only mixture is $\sum_i K_{h_N}(x - X_i)Y_i / \sum_i K_{h_N}(x - X_i)$, the Nadaraya–Watson estimate with a Gaussian kernel. The Gaussian kernel is *regular* (it is bounded below by a positive constant on a ball and $\int \sup_{\|u-x\|\le r} K(u)\,dx < \infty$), and for regular kernels and bounded Y this estimate is strongly consistent in $L_1(\mu)$ under $h_N \to 0$, $N h_N^d \to \infty$ [4]; see Györfi et al. [7, Ch. 5] for the boxed-kernel version in L_2. □

Remark 19 (What the structure is for). Theorem 18 holds because Cobweb never forgets an instance; the hierarchy plays no role in it. The structure matters for three things the theorem does not cover: (a) a *budget* $M < \infty$, where best-first search must find the leaves near the query, (b) the PATH weighting, where the mixture weight of a leaf is its path posterior times its likelihood (Lemma 8 bounds its error), and (c) the *labels* a tree passes to other trees, which are its concepts. The branch-and-bound stop of `categorize(use_best=True)` assumes $\ell_c \le 0$, which holds for nominal attributes but not for Gaussian densities, so with a continuous block it can stop before the best concept.

4.3 Forests learned by Cobweb

The difficulty of learning a forest without gradients is that an inner tree has no target for its own output. In a Cobweb forest each tree models a *joint* distribution, and what it passes on is determined by what it models: an unsupervised tree passes its concepts (labels); a tree that also models y passes its estimate of $\mathbb{E}[y \mid \text{its inputs}]$ (activation). Whether that is enough depends on a property of f relative to the forest's graph.

Definition 20 (Sufficiency)**.** For a forest whose trees all model y (activation messages), let $m_\tau(x) = \mathbb{E}[Y \mid X_{R(\tau)} = x_{R(\tau)}]$, where $R(\tau)$ is the set of raw coordinates upstream of τ. The forest is *sufficient* for the distribution of (X, Y) if, for every tree τ with parents $\sigma_1, \dots, \sigma_k$ and raw inputs x_{S_τ},

$$m_\tau(X) = \psi_\tau\big(m_{\sigma_1}(X), \dots, m_{\sigma_k}(X), X_{S_\tau}\big) \quad \text{a.s. for some continuous } \psi_\tau,$$

i.e. the parents' conditional expectations are sufficient for Y given everything upstream of τ.

Theorem 21 (Consistency of Cobweb-learned forests)**.** *Let X have a continuous density bounded away from zero on $[0,1]^n$, $Y = f(X) + \xi$ with f continuous and ξ bounded, independent, centred noise. Let $\mathfrak{F}$ be a sufficient activation forest of fan-in at most k, learned with the split protocol on N samples, each tree predicting with the leaves-only* LIK *mixture and variance floor $h_{\tau,N}^2$. Assume each ψ_τ is continuous on a neighbourhood of the range of its arguments and the bandwidths satisfy $h_{\tau,N} \to 0$, $N h_{\tau,N}^k / \log N \to \infty$, and $h_{\tau,N} \gg \varepsilon_{\sigma,N}$ for every parent σ, where $\varepsilon_{\sigma,N}$ is the parent's uniform error. Then $\sup_x |\hat{F}_N(x) - f(x)| \to 0$ in probability. If every ψ_τ is Lipschitz, the error is $O_P\big((N/\log N)^{-1/(k+2)}\big)$ up to the composition constants, a rate set by the fan-in k and not by n.*

Proof sketch. Induction in topological order. A source tree τ is a learned tree over (X_{S_τ}, Y); by Lemma 17 its leaves-only LIK activation is the Nadaraya–Watson estimate of m_τ, which converges uniformly on compacts where the design density is bounded below [3, 7]. For a tree τ with parents, the split protocol trains it on a fresh chunk, on the inputs $\hat{U} = (\hat{m}_{\sigma_j}(X), X_{S_\tau})$ instead of $U = (m_{\sigma_j}(X), X_{S_\tau})$; since the Gaussian kernel is Lipschitz, $|K_h(u - \hat{U}_i) - K_h(u - U_i)| \le C\|\hat{U}_i - U_i\| h^{-1} K_{2h}(u - U_i)$ on the event that the deviation is below h, so the estimate built on $\hat{U}$ differs from the one built on U by $O(\max_\sigma \varepsilon_{\sigma,N}/h_{\tau,N})$, which vanishes by assumption. The estimate built on U is the Nadaraya–Watson estimate of $\mathbb{E}[Y \mid U] = m_\tau = \psi_\tau(U)$ by sufficiency,

consistent as before; at test time its input is again $\hat{U}$, handled by the continuity of ψ_τ. The output tree is the last step, where $m_o = f$. For the rate, the Nadaraya–Watson estimate of a Lipschitz function of k variables has uniform error $O_P((N/\log N)^{-1/(k+2)})$ with the optimal bandwidth, and the composition multiplies it by constants. □

Proposition 22 (A sufficient class)**.** *Let the coordinates of X be independent, and let $f(x) = \Phi\big(\sum_p \varphi_p(x_p)\big)$ with Φ strictly increasing and continuous and φ_p continuous. Then every activation forest whose graph is a tree with the raw coordinates at its leaves (each raw coordinate read by exactly one tree) is sufficient.*

Proof. For an upstream set S, $m_S(x) = \mathbb{E}[\Phi(s_S + Z_{\bar{S}})]$ with $s_S = \sum_{p\in S} \varphi_p(x_p)$ and $Z_{\bar{S}}$ independent of x_S; this is a strictly increasing function H_S of s_S. A tree with children covering disjoint sets $S_1, \dots, S_k$ (and raw inputs S_0) has $s_S = \sum_j H_{S_j}^{-1}(m_{S_j}) + \sum_{p\in S_0} \varphi_p(x_p)$, so $m_S = H_S(\cdots)$ is a continuous function of the children's messages and its raw inputs. □

The KAN examples $\exp(\sin \pi x_1 + x_2^2)$, $\exp\big(\frac{1}{100}\sum_i \sin^2(\pi x_i/2)\big)$ and $\exp\big(\frac{1}{2}(\sin \pi(x_1^2 + x_2^2) + \sin \pi(x_3^2 + x_4^2))\big)$ (the last for the graph whose leaves are the pairs (x_1, x_2) and (x_3, x_4)) are in this class.

Proposition 23 (Activations alone can fail)**.** *Let X_1, X_2 be independent and symmetric about 0, and $f(x) = x_1 x_2$. In any activation forest whose output tree reads the activations of two trees over x_1 and x_2 respectively, $m_{\sigma_1} = m_{\sigma_2} \equiv 0$, so the forest's output converges to $\mathbb{E}f = 0$ and its error to $\mathrm{Var}(f)^{1/2}$ for every N and every tree structure.*

Proof. $\mathbb{E}[X_1 X_2 \mid X_1] = X_1 \mathbb{E}X_2 = 0$; by Theorem 18 the source trees' activations converge to 0, so the output tree's input carries no information about x. □

Labels repair this: an unsupervised tree's labels do not depend on f at all.

Theorem 24 (Consistency of label forests)**.** *Let a forest consist of unsupervised trees over the raw coordinates passing labels at depth k_N (or leaf labels) to an output tree that models (labels, Y), learned with the split protocol, and let the output tree predict with the leaves-only LIK mixture with $\alpha_N \to 0$ fast enough ($\alpha_N \cdot \#\text{labels per tree} \cdot n \to 0$). Let $\mathcal{P}_N$ be the product partition of $[0,1]^n$ induced by the frozen label maps. If (a) the diameter of the cell of $\mathcal{P}_N$ containing X tends to 0 in probability and (b) the number of cells of $\mathcal{P}_N$ that contain a sample of the output chunk is $o(N)$, then the forest is a consistent regression estimate.*

Proof sketch. As in the proof of Theorem 15, with $\alpha_N \to 0$ the output tree's leaves-only mixture for a query in cell B puts weight $1 - o(1)$ uniformly on the leaves whose label tuple equals that of B (one per training point of the output chunk in B, by Lemma 17); their y-means average to the partitioning estimate $\sum_{X_i\in B} Y_i/\#\{X_i \in B\}$. Conditional on the frozen partition this is the classical partitioning estimate, consistent under (a) and (b) [7, Thm. 4.2]. □

Remark 25 (The trade-off). Condition (b) requires the product partition to have fewer occupied cells than samples; with K labels per tree that is $K^n = o(N)$, so the rate of a label forest is governed by n, not by the fan-in. Activation forests have rates governed by the fan-in (Theorem 21) but require sufficiency (Proposition 23). Hybrid messages carry both.

Conjecture 26 (Hybrid messages)**.** *A forest passing labels and activations is consistent for every continuous f (through its labels) and attains the fan-in rate of Theorem 21 when the forest is sufficient (through its activations).*

5 Experiments

(Filled in from `results/`; see `experiments/README.md`.)

Claim	Test	Prediction
Theorem 9, (3)	$\sin(\pi x)$, $J_0(20x)$, one tree	error ↓ with N; slope $\approx -1/d$ to $-2/d$
Proposition 10	same, four weightings	PRIOR error plateaus
Theorem 12, Remark 14	4-D and 100-D KAN functions	forest slope set by fan-in; single tree b
Proposition 22, Theorem 21	$\exp(\sin \pi x + y^2)$, 100-D	activation forest consistent
Proposition 23	xy	activation forest error = sd(f); labels/
Theorem 24	label forests, cell diameters	cells shrink; error ↓, slowly
Lemma 17	continual 5-peak task (KAN Fig. 3.4)	no forgetting
—	Feynman equations, tabular, MNIST, text	forest vs single tree vs MLP/KAN

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